KL Divergence, VAE, and ELBO

From surprise and extra bits to the Gaussian closed form and ELBO

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Step-by-step lecture slides

Roadmap

01 Surprise, entropy, and extra bits

02 Closed-form Gaussian KL from mean & variance

03 Why a VAE uses this KL

04 ELBO, step by step

05 Training: reconstruction + KL

What is KL divergence?

Two distributions P (nature) and Q (model) on the same space:

$$\mathrm{KL}(P\|Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)} \qquad \mathrm{KL}(P\|Q) = \int p(x) \log \frac{p(x)}{q(x)}\, dx$$

Non-negative
≥ 0, and $= 0$ iff $P = Q$

Asymmetric
$\mathrm{KL}(P\|Q) \neq \mathrm{KL}(Q\|P)$

Surprise gap
Large where P lives and Q is near 0

Read $\mathrm{KL}(P\|Q)$ *as: extra cost of using* Q *when samples actually come from* P.

Entropy + cross-entropy

$$\mathrm{KL}(P\|Q) = H(P,Q) - H(P)$$

Entropy $H(P)$

$H(P) = -\sum P(x) \log P(x)$

Average surprise of P itself. Optimal code length.

Cross-entropy $H(P,Q)$

$H(P,Q) = -\sum P(x) \log Q(x)$

Code length when we encode P's data with Q.

$$\mathrm{KL}(P\|Q) \;=\; H(P,Q) \;-\; H(P) \;=\; \text{extra bits wasted}$$

Surprise — a weather forecast

$$I(x) = \log \frac{1}{P(x)} = -\log P(x)$$

Independent events multiply; logs make surprise additive.

Fair forecast

sun 50%, rain 50%

$I(\text{sun}) = I(\text{rain}) = \log_2 2 = \mathbf{1\ bit}$

Sunny forecast

sun 90%, rain 10%

$I(\text{rain}) = \log_2 10 \approx \mathbf{3.32\ bits}$

Rain is far more surprising.

If rain is certain ($P = 1$), then $I = 0$: no surprise, no information.

$I = \log_2(1) = 0$

Entropy — average surprise

$$H(P) = \mathbb{E}_{x\sim P}[I(x)] = -\sum_{x} P(x)\log P(x)$$

Fair coin

$H = 1$ bit — one yes/no question.

Certain rain

$H = 0$ — no uncertainty.

Three-way $P = (0.5, 0.25, 0.25)$

sun / cloud / rain

$H(P) = 0.5 \cdot 1 + 0.25 \cdot 2 + 0.25 \cdot 2$

$= \mathbf{1.5}$ bits

A code matched to P uses a short label for sun, longer labels for cloud and rain.

Cross-entropy vs KL — extra bits

True weather $P = (0.5, 0.25, 0.25)$. Wrong model $Q = (0.25, 0.5, 0.25)$ (thinks cloud is most likely).

Entropy	Cross-entropy	KL wasted
1.50	1.75	0.25
bits, true P	bits with Q's code	bits $= 1.75 - 1.50$

$$H(P, Q) = -0.5 \log_2 0.25 - 0.25 \log_2 0.5 - 0.25 \log_2 0.25 = 1.75$$

P is nature. Q is the model. KL is the extra cost of trusting Q.

Discrete calculation (nats)

$$P = (0.7, 0.2, 0.1), \qquad Q = (0.5, 0.3, 0.2) \qquad (\log = \ln)$$

Term	$\log(P_i/Q_i)$	$\times P_i$	Contribution
$i = 1$	0.3365	0.7	$+0.2356$
$i = 2$	-0.4055	0.2	-0.0811
$i = 3$	-0.6931	0.1	-0.0693
Sum $= \mathrm{KL}(P\|Q)$			0.0852 **nats**

Positive $\Rightarrow$ Q is slightly off. Same recipe as weather, different log base.

Forward vs reverse KL

$\mathrm{KL}(P\|Q) \neq \mathrm{KL}(Q\|P)$. The first argument is the reference.

Forward $\mathrm{KL}(P\|Q)$

Mode covering

Punishes Q for missing mass of P. A single Gaussian Q **spreads** to cover both peaks of a bimodal P.

Reverse $\mathrm{KL}(Q\|P)$

Mode seeking

Punishes Q for mass where P is near 0. Q **sits on one peak** and ignores the rest.

VAE uses $\mathrm{KL}\big(q_\phi(\mathbf{z} \mid \mathbf{x}) \,\|\, p(\mathbf{z})\big)$: the encoder must not wander where the prior forbids.

Setup — two 1-D Gaussians

$$q(z) = \mathcal{N}(\mu, \sigma^2) \qquad p(z) = \mathcal{N}(0, 1)$$

$$q(z) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(z-\mu)^2}{2\sigma^2}\right), \quad p(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right)$$

Why a closed form?

A VAE evaluates this KL on every batch. We want μ and σ^2 only — no leftover integral.

Closed form — Step 1: log densities

$$\log q(z) = -\tfrac{1}{2}\log(2\pi\sigma^2) - \frac{(z-\mu)^2}{2\sigma^2}$$

$$\log p(z) = -\tfrac{1}{2}\log(2\pi) - \frac{z^2}{2}$$

$$\mathrm{KL}(q\|p) = \mathbb{E}_{z\sim q}\big[\log q(z) - \log p(z)\big]$$

After cancellation of $\log 2\pi$:

$$\mathrm{KL}(q\|p) = \mathbb{E}_q\left[-\tfrac{1}{2}\log\sigma^2 - \frac{(z-\mu)^2}{2\sigma^2} + \frac{z^2}{2}\right]$$

Closed form — Step 2: moments of q

Under $z \sim \mathcal{N}(\mu, \sigma^2)$:

$\mathbb{E}[z] = \mu$

$\mathbb{E}[(z-\mu)^2] = \sigma^2$

$\mathbb{E}[z^2] = \sigma^2 + \mu^2$

$$\begin{aligned} \mathrm{KL} &= -\tfrac{1}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\underbrace{\mathbb{E}[(z-\mu)^2]}_{\sigma^2} + \tfrac{1}{2}\underbrace{\mathbb{E}[z^2]}_{\sigma^2+\mu^2} \\ &= -\tfrac{1}{2}\log\sigma^2 - \tfrac{1}{2} + \tfrac{1}{2}(\sigma^2+\mu^2) \end{aligned}$$

Closed form — Step 3: mean and variance

$$\mathrm{KL}\big(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0,1)\big) \;=\; \tfrac{1}{2}\big(\mu^2 \,+\, \sigma^2 \,-\, 1 \,-\, \log \sigma^2\big)$$

μ^2

Penalty for moving the latent mean off 0

$\sigma^2 - \log \sigma^2$

Penalty for variance $\neq 1$

-1

Offset so $\mu = 0, \sigma^2 = 1$ gives $\mathrm{KL} = 0$

Check. $\mu = 0$, $\sigma^2 = 1 \Rightarrow \frac{1}{2}(0 + 1 - 1 - 0) = 0$. Encoder matches the prior.

General Gaussians and d dimensions

$$\mathrm{KL}(\mathcal{N}(\mu_1, \sigma_1^2) \,\|\, \mathcal{N}(\mu_2, \sigma_2^2)) = \log \frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}$$

VAE prior $\mu_2 = 0$, $\sigma_2 = 1$ recovers the previous box.

Diagonal d-D latents (standard VAE): KL **adds** over coordinates

$$\mathrm{KL} = \frac{1}{2} \sum_{j=1}^{d} (\mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2)$$

Worked Gaussian examples

Mean shift only

$q = \mathcal{N}(0.2, 1)$, $p = \mathcal{N}(0, 1)$

$\mathrm{KL} = \frac{1}{2}\big(0.04 + 1 - 1 - 0\big)$

$= \mathbf{0.02}$

Variance matches the prior.

Too-narrow encoder

$q = \mathcal{N}(0, 0.25)$, $p = \mathcal{N}(0, 1)$

$\mathrm{KL} = \frac{1}{2}\big(0.25 - 1 - \log 0.25\big)$

$\approx \mathbf{0.318}$

Tiny σ is expensive under the prior.

Pick Gaussians so the analytic KL is known — then you can sanity-check any sampler.

Closed form vs Monte Carlo

$$\mathrm{KL}(q\|p) = \mathbb{E}_{z\sim q}[\log q(z) - \log p(z)]$$

Monte Carlo

Draw $z^{(i)} \sim q$, average log-ratios.

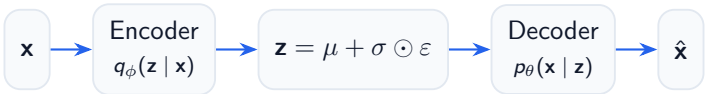
Unbiased but high variance; estimates can even go **negative**.

VAE choice

Plug μ, σ^2 into the closed form.

Cheap, exact (given μ, σ), no leftover sampling noise.

Why a VAE needs KL



Without KL

Isolated tiny codes. Reconstruction works; samples from $\mathcal{N}(0, I)$ do not.

With KL

q_ϕ stays near $p(\mathbf{z}) = \mathcal{N}(0, I)$. Generation from the prior is meaningful.

Target — maximize data likelihood

$$\log p_\theta(\mathbf{x}) = \log \int p_\theta(\mathbf{x} \mid \mathbf{z})\, p(\mathbf{z})\, d\mathbf{z}$$

Problem

This integral is intractable for neural decoders (we cannot sum over all $\mathbf{z}$).

Idea. Introduce $q_\phi(\mathbf{z} \mid \mathbf{x})$ and derive a **tractable lower bound**: the ELBO.

ELBO — Step 1: rewrite with q_ϕ

Insert $\int q_\phi(\mathbf{z} \mid \mathbf{x})\, d\mathbf{z} = 1$ and use $p(\mathbf{x}) = p(\mathbf{x}, \mathbf{z}) / p(\mathbf{z} \mid \mathbf{x})$:

$$\log p_\theta(\mathbf{x}) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_\theta(\mathbf{x}, \mathbf{z})}{p_\theta(\mathbf{z} \mid \mathbf{x})}\right]$$

Meaning

Still **exact**. No approximation yet — only an expectation under the encoder.

ELBO — Step 2: split, expose KL

$$\log p_\theta(\mathbf{x}) = \mathbb{E}_{q_\phi}\left[\log \frac{p_\theta(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z} \mid \mathbf{x})}\right] + \mathrm{KL}\big(q_\phi(\mathbf{z} \mid \mathbf{x}) \,\|\, p_\theta(\mathbf{z} \mid \mathbf{x})\big)$$

$$\log p_\theta(\mathbf{x}) \quad = \quad \mathrm{ELBO}(\theta, \phi; \mathbf{x}) \; + \; \underbrace{\mathrm{KL}(q_\phi \| p_\theta(\mathbf{z} \mid \mathbf{x}))}_{\geq 0}$$

The gap is how far the encoder is from the true posterior.

ELBO — Step 3: the bound

$$\log p_\theta(\mathbf{x}) \qquad \geq \qquad \mathrm{ELBO}(\theta, \phi; \mathbf{x})$$

Floor
ELBO sits under the true log-likelihood

Train
Raising ELBO raises a bound on $\log p(\mathbf{x})$

Tight
Equality iff $q_\phi = p_\theta(\mathbf{z} \mid \mathbf{x})$

ELBO — Step 4: VAE training form

$$\mathrm{ELBO} = \underbrace{\mathbb{E}_{q_\phi}[\log p_\theta(\mathbf{x} \mid \mathbf{z})]}_{\text{reconstruction}} - \underbrace{\mathrm{KL}\big(q_\phi(\mathbf{z} \mid \mathbf{x}) \,\|\, p(\mathbf{z})\big)}_{\text{KL regularizer}}$$

Reconstruction

Can the decoder rebuild $\mathbf{x}$ from $\mathbf{z}$?
Often $-\mathrm{MSE}$ or $-\mathrm{BCE}$.

KL regularizer

Does the encoder stay near $\mathcal{N}(0, I)$?
Use the closed form we derived.

Practical training loss

We **minimize** $-\mathrm{ELBO}$:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta \cdot \mathrm{KL}\big(q_\phi(\mathbf{z} \mid \mathbf{x}) \,\|\, \mathcal{N}(0, I)\big)$$

$\beta = 1$	$\beta > 1$	$\beta \to 0$
Standard VAE	Stronger structure	Plain autoencoder

Reparameterization

Sampling $\mathbf{z} \sim q_\phi$ is stochastic. Push noise into ε:

$$\varepsilon \sim \mathcal{N}(0, I), \qquad \mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}) + \boldsymbol{\sigma}_\phi(\mathbf{x}) \odot \varepsilon$$

ε
No parameters

μ, σ
Deterministic in $\mathbf{x}$

Gradients
Flow into encoder ϕ

One training step

1. Encode $\mathbf{x} \to (\boldsymbol{\mu}, \boldsymbol{\sigma})$
2. Sample $\varepsilon \sim \mathcal{N}(0, I)$; $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \varepsilon$
3. Decode $\mathbf{z} \to \hat{\mathbf{x}}$; compute $\mathcal{L}_{\text{recon}}$
4. $\mathrm{KL} = \frac{1}{2} \sum_j (\mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2)$
5. $\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta\,\mathrm{KL}$; backprop; update θ, ϕ

Numerical ELBO — 1-D latent

One point: $\mu = 0.5$, $\sigma^2 = 0.8$, $\mathcal{L}_{\text{recon}} = 0.40$

KL

$\frac{1}{2}(0.25 + 0.8 - 1 - \log 0.8)$
$= \frac{1}{2}(0.27314) \approx \mathbf{0.1366}$

Loss / ELBO

$\mathcal{L} \approx 0.40 + 0.1366 = \mathbf{0.5366}$
$\text{ELBO} \approx -0.5366$

Higher ELBO = tighter (better) bound on $\log p(\mathbf{x})$.

Takeaways

1. KL = extra bits of using model Q when nature is P. Always ≥ 0.
2. Closed form: $\frac{1}{2}(\mu^2 + \sigma^2 - 1 - \log \sigma^2)$ from $\mathbb{E}[(z-\mu)^2] = \sigma^2$ and $\mathbb{E}[z^2] = \sigma^2 + \mu^2$.
3. VAE uses that KL so latents stay near $\mathcal{N}(0, I)$.
4. $\log p(\mathbf{x}) = \mathrm{ELBO} + \mathrm{KL}(q \| p(\mathbf{z} \mid \mathbf{x}))$.
5. Train: reconstruction quality minus KL; reparameterize $\mathbf{z}$.

Practice. Show $\mathrm{KL} = 0$ when $\mu = \mathbf{0}$, $\sigma = \mathbf{1}$. What does that say about the encoder?